

# OS Banana - 2 Assignment

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Q1)

A

```
[krish@archlinux PES1UG24AM027-pes-vcs]$ make test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
[krish@archlinux PES1UG24AM027-pes-vcs]$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check
```

B

```
All Phase 1 tests passed.
[krish@archlinux PES1UG24AM027-pes-vcs]$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
[krish@archlinux PES1UG24AM027-pes-vcs]$ █
```

Q2)

A

```
[krish@archlinux PEs1UG24AM027-pes-vcs]$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
[krish@archlinux PEs1UG24AM027-pes-vcs]$
```

B

```
[krish@archlinux PEs1UG24AM027-pes-vcs]$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
[krish@archlinux PEs1UG24AM027-pes-vcs]$ xxd .pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
00000000: 626c 6f62 2036 0068 656c 6c6f 0a          blob 6.hello.
[krish@archlinux PEs1UG24AM027-pes-vcs]$
```

Q3)

```
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
cat .pes/index    # Human-readable text format
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
  untracked: .gitignore
  untracked: tree.h
  untracked: tree.c
  untracked: test_tree.c
  untracked: test_sequence.sh
  untracked: test_objects.c
  untracked: README.md
  untracked: pes.h
  untracked: pes.c
  untracked: object.h
  untracked: object.c
  untracked: Makefile
  untracked: index.h
  untracked: index.c
  untracked: commit.h
  untracked: commit.c

100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776748274 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776748274 6 file2.txt
[krish@archlinux PES1UG24AM027-pes-vcs]$
```

Q4)

A

```
./pes log
Initialized empty PES repository in .pes/
[6366c9cf2b0a5620c1b0298873d5aaa362bcc147d27172771f1ab146734b9851] Initial commit
Committed: 6366c9cf2b0a... Initial commit
[ca47fbd39f73afb0ca275746ca9ed8011c2942bb741cd92762e5b9717cbdda52] Add world
Committed: ca47fbd39f73... Add world
[48539e8b3b3ee326825807cc16e5bbda79be782f6fe2f6f1a3c7e04634d6cc4f] Add farewell
Committed: 48539e8b3b3e... Add farewell
commit 48539e8b3b3ee326825807cc16e5bbda79be782f6fe2f6f1a3c7e04634d6cc4f
Author: Your Name <PES1UG24AM027>
Date: 1776748860

    Add farewell

commit ca47fbd39f73afb0ca275746ca9ed8011c2942bb741cd92762e5b9717cbdda52
Author: Your Name <PES1UG24AM027>
Date: 1776748860

    Add world

commit 6366c9cf2b0a5620c1b0298873d5aaa362bcc147d27172771f1ab146734b9851
Author: Your Name <PES1UG24AM027>
Date: 1776748860

    Initial commit

[krish@archlinux PES1UG24AM027-pes-vcs]$
```

B

```
[krish@archlinux PES1UG24AM027-pes-vcs]$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/48/539e8b3b3ee326825807cc16e5bbda79be782f6fe2f6f1a3c7e04634d6cc4f
.pes/objects/63/66c9cf2b0a5620c1b0298873d5aaa362bcc147d27172771f1ab146734b9851
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/ca/47fbd39f73afb0ca275746ca9ed8011c2942bb741cd92762e5b9717cbdda52
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
[krish@archlinux PES1UG24AM027-pes-vcs]$
```

C

```
[krish@archlinux PEs1UG24AM027-pes-vcs]$ cat .pes/refs/heads/main  
48539e8b3b3ee326825807cc16e5bbda79be782f6fe2f6f1a3c7e04634d6cc4f  
[krish@archlinux PEs1UG24AM027-pes-vcs]$ cat .pes/HEAD  
ref: refs/heads/main  
[krish@archlinux PEs1UG24AM027-pes-vcs]$
```